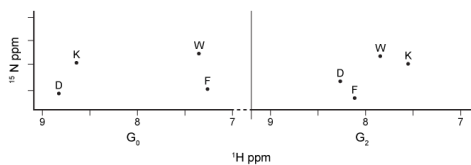


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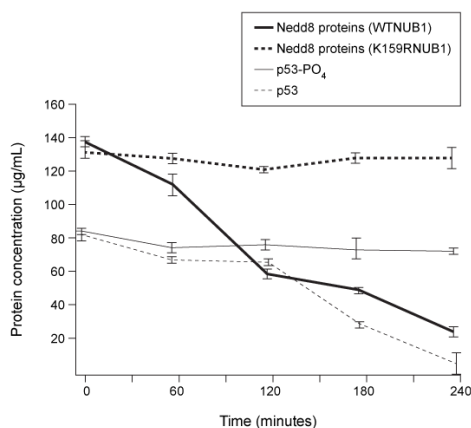
**Figure 1** Organization of the p53 domains (1–393 amino acids)

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**Figure 3** Half-lives of p53, phosphorylated p53 (p53- $\text{PO}_4$ ), and Nedd8 proteins in cells with wild-type or mutant NUB1 over time

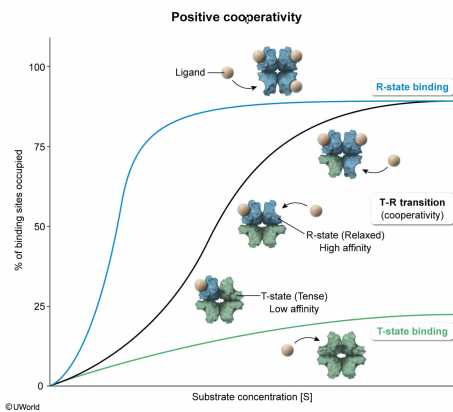
p53 displays positive cooperativity when it binds to promoters on DNA. Which statement regarding p53 structure and function accurately explains this observation?

- ☐ A. The DBD subunits bind individually to DNA and have independent binding affinities. [1%]
- ☐ B. Phosphorylation of the REG increases the affinity of the DBD for DNA. [4%]
- ☒ C. The affinity of one DBD subunit for DNA increases as adjacent subunits bind to the DNA. [91%]
- ☐ D. Once the first promoter is bound, the binding affinity of the DBD decreases. [2%]

Correct

91% answered correctly

## Explanation:



Proteins that change their shape or conformation when a substrate or ligand binds to them are referred to as **allosteric proteins**. Some proteins have quaternary structure, meaning they are composed of multiple polypeptide chains known as subunits, which allows for a special kind of allosteric behavior known as **cooperativity**. In cooperative proteins, the binding of one subunit induces a **conformational change** in adjacent subunits, which results in a new protein shape with a higher (R-state) or lower (T-state) affinity for the substrate or ligand. **Cooperativity** can be positive or negative: proteins with subunits that increase in affinity when other subunits are bound are said to exhibit **positive cooperativity** whereas those that lose affinity exhibit **negative cooperativity**. Proteins that do not change affinity are noncooperative.

p53 is a homotetrameric (ie, four identical subunits) protein that displays positive cooperativity for DNA. Therefore, the binding affinity of the DBD in each subunit increases after one subunit has bound to the DNA.

**(Choice A)** If the affinity of each subunit were independent of the binding states of other subunits, p53 would be noncooperative.

**(Choice B)** Post-translational modifications such as phosphorylation can induce conformational changes but are not involved in cooperativity because phosphorylation is a covalent attachment, not a binding event.

**(Choice D)** A decrease in binding affinity of the DBD of one subunit when another subunit binds to them would be an example of negative cooperativity.

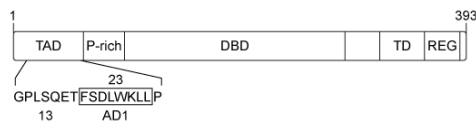
## Educational objective:

Cooperativity refers to the conformational changes in one subunit of a protein that increase or decrease the binding affinity of the other subunits. Ligand/substrate binding that increases the binding affinity of other subunits is referred to as positive cooperativity, and binding that decreases affinity is called negative cooperativity.

**Time spent:** 0 sec  
**Subject:**  
Biochemistry

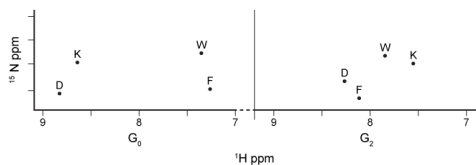
**QID:** 400986  
**System:**  
Amino Acids and Proteins

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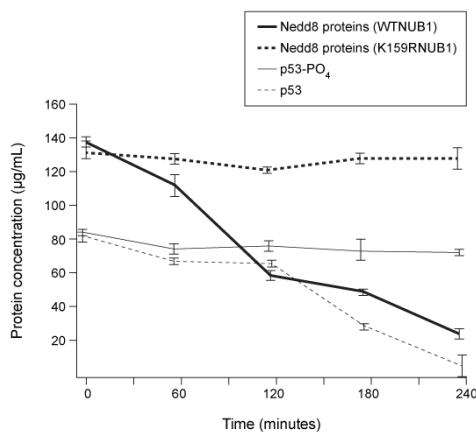
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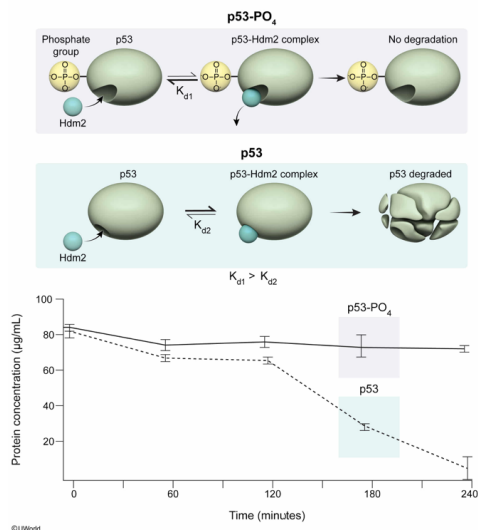
The data in Figure 3 support which of the following conclusions?

- ☒ A. The high  $K_d$  of the Hdm2-p53- $\text{PO}_4$  complex causes p53- $\text{PO}_4$  concentrations to remain steady. [65%]
- ☐ B. Unphosphorylated p53 is degraded slowly because it has a low affinity for Hdm2. [15%]
- ☐ C. Dephosphorylation reduces the p53 degradation rate by increasing the  $K_d$  of Hdm2-p53- $\text{PO}_4$ . [11%]
- ☐ D. Phosphorylated p53 is prone to cleavage due to its greater affinity for Hdm2. [7%]

Correct

64% answered correctly

## Explanation:



**Binding affinity** describes the strength of noncovalent binding interactions between two molecules. The **equilibrium dissociation constant**  $K_d$  indicates the tendency of a protein-ligand complex to separate into free protein and free ligand. A **low  $K_d$  value** is indicative of a high tendency for the complex to remain bound, and therefore indicates a **high binding affinity**; alternatively, a high  $K_d$  value suggests a low binding affinity. In other words, the dissociation constant of a protein-ligand complex is inversely proportional to its **binding affinity**.

Regulatory proteins such as Hdm2 can **negatively** or positively affect the activity, half-life, or expression of the proteins to which they bind. Based on the data in Figure 3, Hdm2 can be classified as a **negative regulator** of p53 because it binds and signals for the cleavage or degradation of unphosphorylated p53. The half-life graph in Figure 3 illustrates that **unphosphorylated p53 proteins have a much shorter half-life than their phosphorylated counterparts**. This difference in half-lives reflects the gap between the dissociation constants of phosphorylated and unphosphorylated p53.

The passage suggests that **phosphorylation of p53 at residue T18 reduces the binding affinity** of p53 for Hdm2. Consequently, phosphorylated p53 is **degraded at a slower rate** because it has a **greater  $K_d$**  (ie, lower affinity for Hdm2) than unphosphorylated p53.

**(Choice B)** Figure 3 shows that unphosphorylated p53 is degraded rapidly. The passage states that Hdm2 binds p53, indicating a low  $K_d$ .

**(Choice C)** Dephosphorylation of p53 increases the rate of p53 degradation by decreasing  $K_d$  (ie, raising the binding affinity) of the Hdm2-p53-PO<sub>4</sub> complex.

**(Choice D)** Phosphorylated p53 is cleaved *less* readily because it has *lower* affinity (ie, greater  $K_d$ ) for Hdm2.

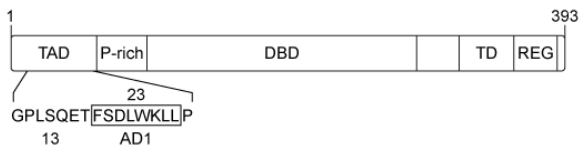
## Educational objective:

Binding affinity describes the strength of noncovalent interactions between two molecules and is indicated using the equilibrium dissociation constant  $K_d$ . The dissociation constant of a protein for its substrate is inversely proportional to its binding affinity. Negative regulators decrease the expression, activity, or half-life of their target proteins.

Time spent: 0 sec  
Subject:  
Biochemistry

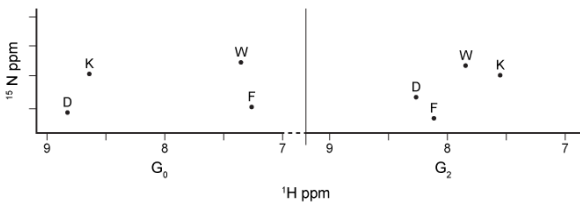
QID: 400987  
System:  
Amino Acids and Proteins

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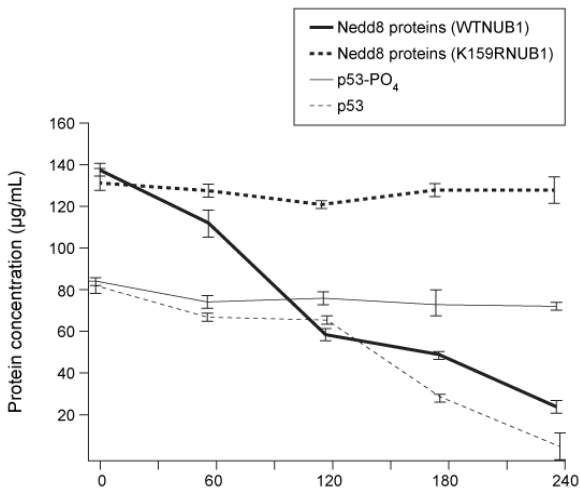
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**Figure 3** Half-lives of p53, phosphorylated p53 (p53-PO<sub>4</sub>), and Nedd8 proteins in cells with wild-type or mutant NUB1 over time

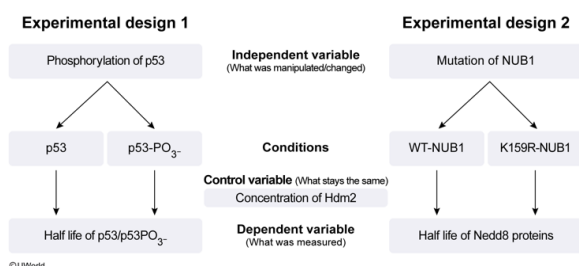
In Figure 3, why was protein concentration measured only in human embryonic kidney cells that overexpressed Hdm2?

- ☐ A. Overexpression of Hdm2 is required for p53 and NUB1 degradation. [10%]
- ☐ B. Protein concentration is dependent on the half-life of Hdm2. [3%]
- ☐ C. It provides a baseline for the comparison of Nedd8 and p53 levels. [17%]
- ✓ ☒ D. It ensures Hdm2 concentration was not a limiting factor for degradation. [69%]

Correct

69% answered correctly

### Explanation:



The goal of the study in the passage was to determine the effect both of phosphorylation on the half-life of p53 and of the influence of a K159R mutation on the activity of NUB1 in the presence of Hdm2. To test these relationships, two experiments were conducted involving either p53 or NUB1. The phosphorylation state of p53 and the presence of WTNUB1 or K159RNUB1 were the **independent variables** controlled in each experiment, and the concentration of p53 or Nedd8 proteins served as the **dependent variables**.

**Limiting factors** restrict or control the rate of reaction by establishing the maximum level at which the reaction proceeds. Cells that overexpressed Hdm2 were used so that an excess of Hdm2 was available to bind p53 and NUB1 proteins, ensuring that the concentrations of p53 and Nedd8 depended only on the phosphorylation state or a mutation and not on Hdm2 concentration. In other words, the presence of excess Hdm2 ensured that it was not a limiting factor for the experiment. Hdm2 concentration acted as a **constant or control variable**, which is a variable that remains unchanged and allows the relationship between independent and dependent **variables** to be accurately tested.

**(Choice A)** NUB1 degradation was not measured in the experiment. Nedd8 protein levels function as markers of NUB1 activity.

**(Choice B)** The experiment measured the concentrations of Nedd8 and p53 proteins. Protein concentration was measured in relation to Hdm2 binding, not half-life.

**(Choice C)** Nedd8 proteins and p53 levels were not compared to each other in the experiment.

**Educational objective:**

Independent variables are the conditions controlled by the experimenter that can affect the dependent variable, or that which is being measured. Constant or control variables remain unchanged during an experiment and allow the accurate interpretation of results. Limiting factors restrict or control the rate of reaction by establishing the maximum level at which the reaction proceeds.

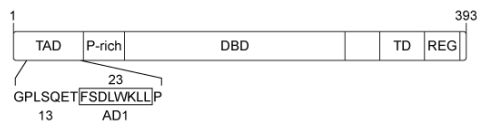
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**Subject:**  
Biochemistry

**QID:**400988

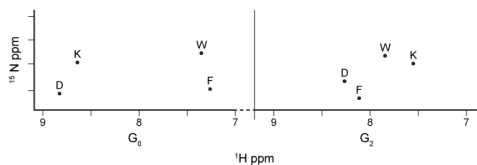
**System:**  
Amino Acids and Proteins

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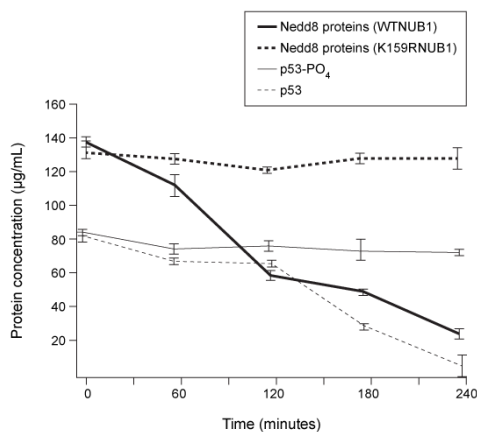
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**Figure 3** Half-lives of p53, phosphorylated p53 (p53- $\text{PO}_4$ ), and Nedd8 proteins in cells with wild-type or mutant NUB1 over time

Which statement best explains the relationship between Hdm2 and Nedd8 protein concentration?

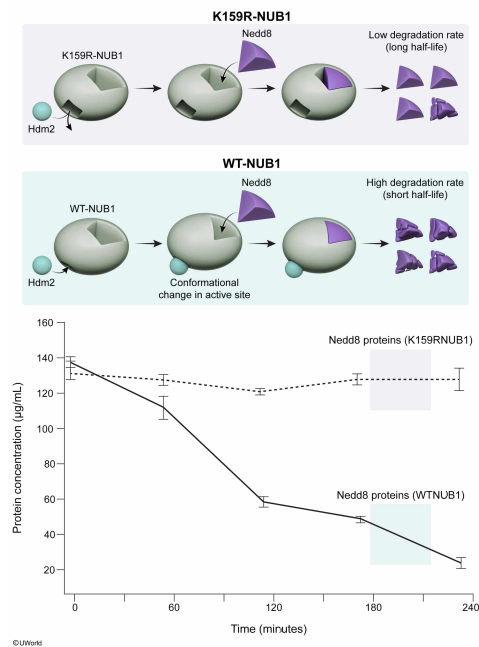
- ☐ A. Nedd8 protein concentration is directly proportional to Hdm2 activity. [7%]
- ☒ B. Degradation of Nedd8 protein is more efficient when Hdm2 associates with NUB1. [73%]
- ☐ C. NUB1-mediated Nedd8 protein degradation is independent of Hdm2 activity. [7%]
- ☐ D. Nedd8 protein levels are only dependent on the ability of Hdm2 to degrade NUB1. [10%]

Correct

74% answered correctly



## Explanation:



**Positive regulators** bind reversibly at positions outside the active site and **enhance the activity** of their target protein. Hdm2 regulates protein activity by attaching a ubiquitin molecule to its target protein. According to the passage, NUB1 proteins are targets for Hdm2 regulation. The function of wild-type NUB1 (WTNUB1) is to enhance signaling for the degradation of Nedd8 proteins, so its presence reduces the half-life of Nedd8.

Based on the data shown in Figure 3, Hdm2 acts as a positive regulator of NUB1 proteins: in the presence of Hdm2, WTNUB1 induces the continuous degradation of Nedd8. However, K159R mutants do not induce the degradation of Nedd8 proteins. This reduction in function implies that the K159R mutation prevents Hdm2 from binding to NUB1. As a result, K159R mutants do not actively reduce the half-life of Nedd8 proteins even though Hdm2 proteins are present. Therefore, the degradation of Nedd8 proteins is more efficient when NUB1 is bound by Hdm2.

**(Choice A)** Hdm2 activity increases Nedd8 degradation; therefore, Nedd8 protein concentration is inversely proportional to Hdm2 activity.

**(Choice C)** Nedd8 protein concentration decreases when WTNUB1 is bound to Hdm2.

**(Choice D)** Hdm2 binding does not signal for the degradation of NUB1 proteins. Instead, Nedd8 protein levels depend on NUB1 being activated by Hdm2.

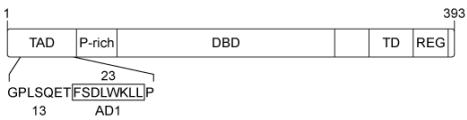
## Educational objective:

Positive regulators bind reversibly at positions outside the active site and enhance the activity of their target proteins.

Time spent: 0 sec  
Subject:  
Biochemistry

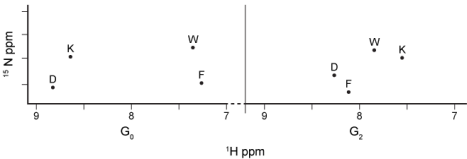
QID: 400989  
System:  
Amino Acids and Proteins

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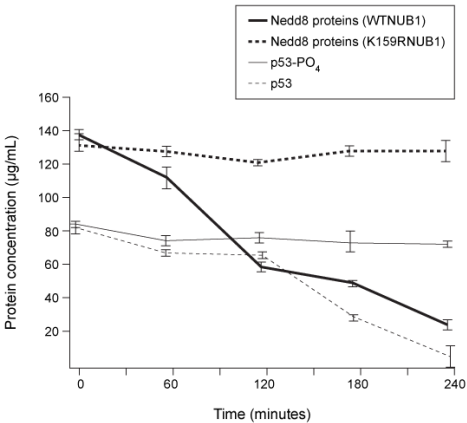
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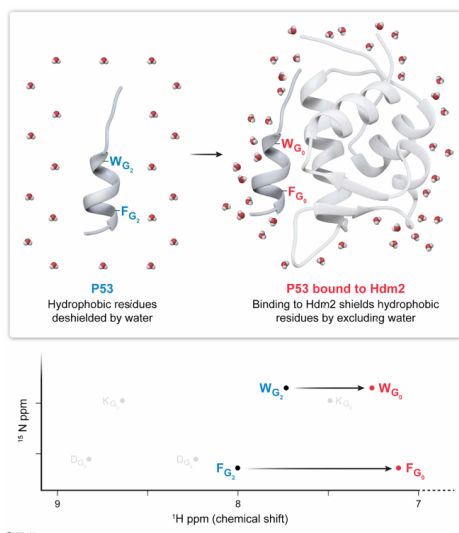
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**Figure 3** Half-lives of p53, phosphorylated p53 (p53-PO<sub>4</sub>), and Nedd8 proteins in cells with wild-type or mutant NUB1 over time

Based on the data in Figure 2, which statement best supports the formation of a p53-Hdm2 complex in the absence of DNA errors?

- ☐ A. Protons on hydrophilic amino acids of p53 are more shielded during G<sub>0</sub>. [16%]
- ☐ B. Protons on hydrophilic amino acids of p53 are deshielded during G<sub>2</sub>. [16%]
- ☒ C. Protons on hydrophobic amino acids of p53 are more shielded during G<sub>0</sub>. [51%]
- ☐ D. Protons on hydrophobic amino acids of p53 are more shielded during G<sub>2</sub>. [15%]

**Explanation:**

Proton NMR spectroscopy measures the frequency of hydrogen nuclei spinning in a magnetic field. This frequency, referred to as the **chemical shift**, is measured in parts per million (ppm) and describes the electron density surrounding a proton within a molecule. When surrounded by **high electron density**, protons are said to be "**shielded**" from the magnetic field. However, **electronegative atoms** draw electron density from nearby protons, causing them to become **deshielded**. Protons that are shielded appear to the right (upfield) of the NMR spectrum whereas deshielded protons appear to the left (downfield).

During protein-protein interactions, hydrophobic residues often **associate** to **avoid contact with water**. This interaction increases the electron density around the hydrophobic residues by shielding them from the electronegative oxygen in water.

The spectrum in Figure 2 was obtained by analyzing protons on the amino acid residues in the binding region of p53. The passage states that p53 binding to Hdm2 is disrupted by errors in replication. Because DNA replication does not occur in  $G_0$  (and therefore errors are not present), the changes in the chemical shift from  $G_0$  to  $G_2$  are likely related to p53 binding to Hdm2. Protons on the hydrophobic amino acids, F and W, are more shielded (ie, shifted to the right) in  $G_0$  than in  $G_2$ , suggesting that hydrophobic interactions are enhanced in  $G_0$ , and therefore that p53 binds Hdm2.

**(Choices A and B)** Hydrophilic amino acids on the surface of p53 are more deshielded (ie, shifted to the left) during  $G_0$  than  $G_2$ .

**(Choice D)** Hydrophobic amino acids are more shielded during  $G_0$ , when p53 is expected to bind Hdm2.

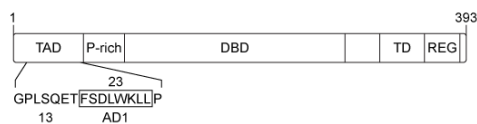
**Educational objective:**

High electron densities protect or "shield" protons (right shift) from the magnetic field of the spectrometer whereas electronegative atoms deshield protons (left shift). Hydrophobic residues associate and shield protons at protein-protein interfaces by avoiding water molecules.

Time spent: 0 sec  
Subject:  
Biochemistry

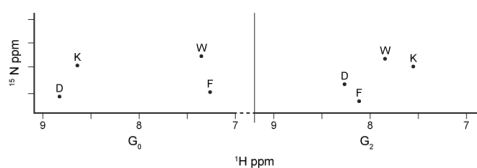
QID: 400990  
System:  
Amino Acids and Proteins

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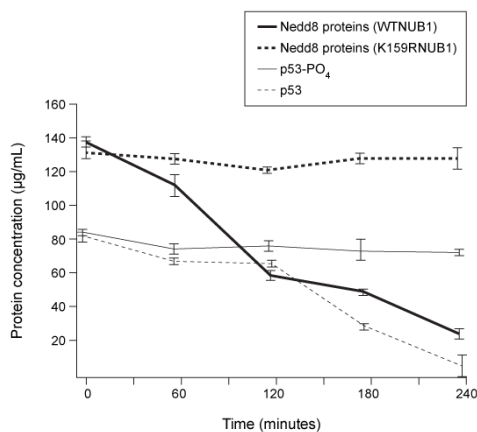
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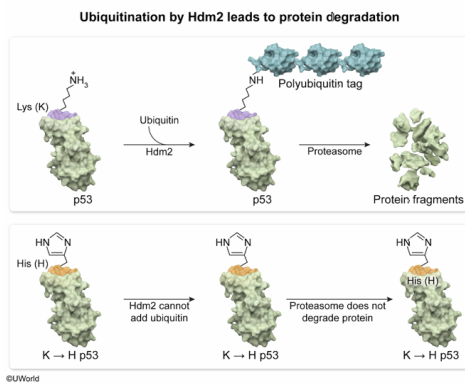
Which of the following mutations would most likely decrease the degradation of p53?

- ☐ A. V  $\rightarrow$  L in the DBD [6%]
- ☒ B. K  $\rightarrow$  H in the TAD [63%]
- ☐ C. T  $\rightarrow$  S in the N-terminal domain [15%]
- ☐ D. P  $\rightarrow$  G in the P-rich domain [14%]

Correct

63% answered correctly

## Explanation:



**Ubiquitin** is a small protein that can be attached to other proteins as a **molecular signal**. It is usually linked to proteins through a lysine side chain, and **polyubiquitination** is a signal that **targets proteins to the proteasome** for degradation.

Small portions of a polypeptide chain can fold into distinct three-dimensional structural units known as **domains**, which independently carry out unique functions. Figure 1 shows that p53 has five different domains that span 393 amino acids. According to the passage, Hdm2 is a lysine-specific ubiquitin ligase that attaches ubiquitin to the **TAD domain** to signal for proteolytic cleavage. Because Hdm2 is lysine-specific, mutating the relevant lysine to any other amino acid interferes with ubiquitination. Therefore, a **K → H mutation** in the TAD domain would most likely **decrease p53 degradation**.

**(Choice A)** Hdm2 does not interact with **aliphatic amino acids** such as valine (V) or leucine (L).

**(Choice C)** Serine (S) and threonine (T) have similar properties, so a T-to-S mutation in the N-terminal domain would likely either have minimal effect or, if the mutation prevented phosphorylation at position 18, increase degradation.

**(Choice D)** A proline (P) to glycine (G) mutation in the P-rich domain is unlikely to alter p53 degradation because no evidence is provided that this domain is involved in signaling for proteolysis.

### Educational objective:

The protein ubiquitin can be attached to other proteins as a molecular signal, usually through linkage to a lysine residue. Polyubiquitination is a signal for protein degradation by the proteasome. Mutating the

ubiquitinated lysine to any other amino acid decreases the rate of protein degradation.

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